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*Geological Society, London, Special Publications* 1998, v.144; p205-220.

doi: 10.1144/GSL.SP.1998.144.01.16

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**Notes**

# Secondary oil migration: oil–rock contact volumes, flow behaviour and rates

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**Abstract:** Determining oil–rock contact volumes associated with secondary migration is a major challenge due to the heterogeneous nature of the sedimentary structures and petrophysical properties of carrier systems at multiple length scales. In this paper, we argue that migration occurs at, or near, the percolation threshold of the carrier system and show that the oil–rock contact volumes are a function of the heterogeneity of the threshold pressure field of the medium, as well as the orientation of the sedimentary fabric relative to the orientation of the net migration vector. The flow properties are intimately tied to the flow mechanism, which for secondary oil migration is likely to occur via capillary creep. An understanding of the distribution of threshold pressures within a sample volume will enable us to estimate the oil–rock contact volume during secondary oil migration, whether the volume considered is a core plug in size or an entire carrier sequence.

Much work has been done on trying to derive gross estimates of petroleum loss during secondary migration. The majority of these methods are based on the premise that the volume of petroleum lost is proportional to the pore volume through which the petroleum has migrated. Using mass balance calculations Mackenzie & Quigley (1988) give two examples where the loss rates along the migration pathway were calculated by subtracting the accumulation volumes from a modelled expulsion volume using an estimated pore volume of the carrier. For both cases, the inferred oil–rock contact was roughly 2% of the total drainage volume.

England *et al.* (1987) suggest that, due to the balance of the viscous and capillary forces combined with the belief that oil migrates as a continuous phase, water is unlikely to displace petroleum in the carrier following the depletion of the oil supply. Based on this, we can expect the residual oil volume to be equivalent to, the volume necessary to create the initial interconnected pathway across the carrier. Expressed in terms of percolation theory, the volume of residual oil will be close to, or equivalent to the volume of the spanning cluster. By extension, therefore, understanding the morphology of the migration path is critical for estimating residual saturation (and migration efficiencies).

It is generally accepted that capillary pressure is the dominant constraint to oil migration (e.g. Berg 1975; Schowalter 1979; England *et al.* 1987; Ringrose & Corbett 1994). The buoyancy force exerted by the density difference between the oil phase and the surrounding pore water will cause the oil stringer to migrate upwards, restricted only by the opposing capillary forces. Migration can therefore be viewed as the oil

‘breaking’ its way through a network of opposing capillary pressures. These breakthrough pressures correspond to the pressure required for a continuous phase to span the distance between the inlet and outlet of a given rock volume. Knowing the phase saturations at breakthrough for various lithotypes, at various length scales, can therefore provide valuable insights into both oil–rock contact volumes and residual saturations during migration.

In this paper, we will first present our view on how oil migrates in non-fractured, clastic carrier sequence. We will then examine each of our assumptions and conclusions against published and numerical modelling results from each of the sub-disciplines used to construct this generalized model. We will show that the precise morphology of the migration pathway is a function of the threshold pressures within the carrier sequence and that these, not permeability, can be used to rapidly assess migration pathway potentials along with their associated oil–rock contact volumes.

## Terminology used in this paper

**Carrier:** The carrier is the permeable matrix through which the oil migrates. In this paper, we focus exclusively on unfractured, unfaulted, clastic carriers, although the techniques and concepts we describe can be extended to other systems.

**Threshold pressure:** The threshold pressure is the applied pressure which is required to move a non-wetting phase (oil) through a given volume of carrier material which has been saturated

with a wetting phase (water). It is the precise pressure at which the non-wetting phase appears at the outlet of the sample volume (i.e. at breakthrough). It is usually synonymous with *breakthrough pressure* and *displacement pressure*.

**Spanning cluster:** At the point of breakthrough, the spanning cluster is defined as the group of pores which collectively span the sample volume to create a continuous stringer of oil. Once the spanning cluster is formed, the system is said to percolate.

**Oil stringer:** An oil stringer represents a continuous phase of oil which exists between two points within a carrier medium. A stringer may migrate a small distance due to its own buoyancy, but requires replenishment if it is to grow and migrate any appreciable distance.

**Ganglion:** Similar to an oil stringer but referring to the pore scale where a volume of oil becomes isolated due to pore-scale snap-off.

**Backbone:** The backbone is the sub-set of the spanning cluster which will have the highest flow velocities when the oil phase is mobile. It is the biased path of least resistance through the sample, which will also corresponds to the volume containing the highest oil saturations.

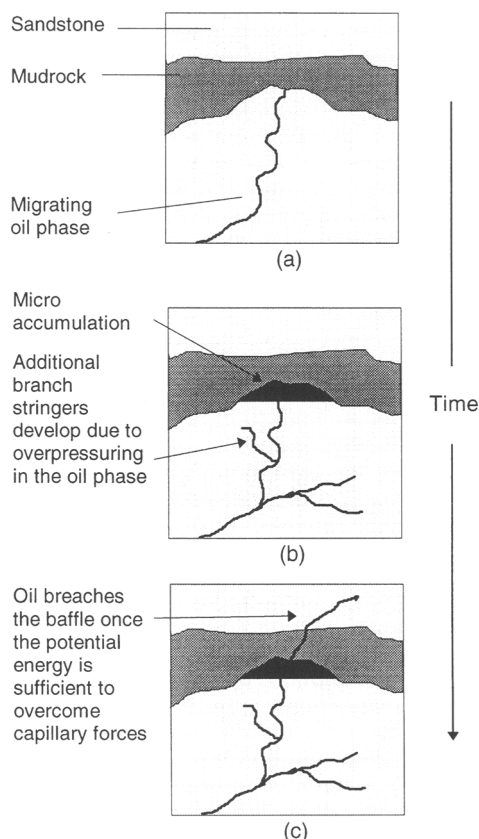
## How oil migrates – a conceptual view

Oil is released into a carrier bed from a presumed source rock at extremely low charge rates relative to oil field production rates. This low rate has the effect of limiting the spread of the oil front, restricting the oil–rock contact to a small percentage of the carrier volume which, as we will show, is ultimately defined by the internal threshold pressure field of the rock volume.

Once a critical saturation is reached, the oil migrates as a separate, immiscible phase, within a water-saturated carrier, continuous at the margins of the oil front. The density difference between the oil phase and the surrounding pore water creates a buoyancy force which is proportional to the difference between the two phases. Restricting its flow is the capillary pressure created at the interface of the oil and water within the largest upstream pore throat. The oil will, in general, migrate vertically under gravity-destabilized conditions according to Equation (1):

$$(\rho_o - \rho_w)gz > P_c. \quad (1)$$

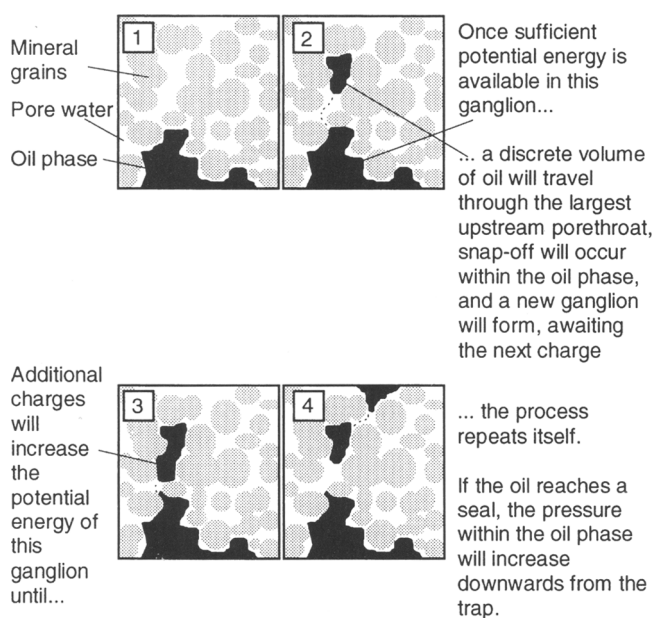
Figure 1a shows a cartoon representation of an oil phase (stringer) migrating through a high permeability carrier, beneath a low permeability



**Fig. 1.** Generalized model of how oil migrates.

baffle. The oil will migrate unimpeded until it reaches the baffle, at which point a micro-accumulation may develop (Fig. 1b). During the build-up of this accumulation (when a no-flow condition exists), the pressure within the oil phase will increase throughout the continuous portion of the phase according to Pascal's principle, which states that when an external force (the feeding of the oil from the source) is applied to a confined body of fluid, the internal pressure increases by the same amount at every point throughout the system (Dahlberg 1995). The oil will remain trapped beneath this seal until sufficient potential energy is available within the oil stringer for the resisting capillary threshold pressure of the baffling unit to be overcome. Depending on the distribution of threshold pressure values of the surrounding medium, the accumulation may develop as branches off the principal migration pathway, creating a more dispersed oil front (as in Fig. 1b).

A critical concern is the continuity of the oil phase. Oil will migrate as a discrete stringer (or



**Fig. 2.** Pore-scale cartoon of oil migration in a water-wet carrier.

ganglion) until halted by a collection of pore throats. It will then only continue if replenished by a subsequent charge of oil (see Fig. 2). Individual ganglia will then slowly migrate upwards, coalesce or split in a complex pore-scale dynamic process. There will be no pressure communication between ganglia. Once a trap is reached, ganglia will coalesce to form a continuous phase beneath the seal or baffle. At some point the accumulation may then migrate as an oil stringer, and so on.

Once the oil has travelled through a given volume of rock, the degree of residual oil left behind will be equal to the differences between the end points on drainage and imbibition relative permeability curves of the lithotype in which the migration occurred. These end point differences are a function of topological considerations within the pore space (e.g. Wardlaw & McKellar 1981).

After breakthrough of a given volume, oil migration will occur along the flow backbone of the medium. Independent of the total potential energy of the oil phase, the backbone represents the path of least capillary resistance through the oil-charged medium. This suggests that if we were able to locate this backbone spatially, we should be able to predict the trajectory of secondary oil migration. In addition, the volume of this spanning network is finite, suggesting that the flow within the backbone

occurs at a single saturation (consistent with England *et al.*'s (1987) earlier claim). It may therefore be appropriate to model migration using a single value for relative permeability for a given rock volume – a value corresponding with the critical oil saturation required to form a spanning cluster across the volume.

This conceptual view of how oil migrates implies that, in terms of oil–rock contact volumes, the initial oil charge will come into contact with a higher volume of rock than successive charges. It is the first charge which will form the spanning cluster. Once this has been achieved, successive charges will migrate exclusively along the established backbone, as we will show below.

There are a number of assumptions in this view on how oil migrates, and it is also one which is difficult to validate, since one cannot examine and inspect the dynamism of migration pathways over the time scale of secondary migration. Various authors (e.g. Dembicki & Anderson 1989; Thomas & Clouse 1995) have attempted to reproduce the process in the laboratory; however, the flow rates used were much higher than would occur in reality. There is, however, geochemical evidence to suggest that this is the way migration occurs (e.g. Larter & Aplin 1995; Rasmussen 1997).

In the following sections, we will examine each of our assumptions in this conceptual view using separate models which, when combined, will

increase our understanding of this phenomenon. To begin, we will examine the effects of feeding rates on the morphology of the oil front and show that it is not permeability which controls the contact volume, but rather threshold pressures. Then we will look more closely at the threshold pressure/oil–rock contact volume relationship and examine the role of both heterogeneous pressure fields and sedimentary architecture on oil–rock contact volumes.

### Feeding rates and oil-front morphology

The feeding rate of the oil phase into a water-wet carrier has a dramatic effect on the form of the migration pathway and, by extension, the oil–rock contact volume. Figure 3 demonstrates this using a 36 m<sup>2</sup> model of an inclined (20°), planar laminated sandstone.

We used a finite difference, black oil simulator (Eclipse) to model migration through the permeability field shown in Fig. 3a. The field has a permeability distribution in the range of 1 to 100 mD. Empirical reservoir engineering relative permeability and capillary pressure functions were used (see Appendix).

Figure 3 shows the oil front invading the sandstone from an injection point located in the lower right corner, at three different injection rates. As the rate is lowered, the oil begins to follow, almost exclusively, the backbone of the system. For the higher flow rates, the total pressure exerted on water-filled pores is the sum of the

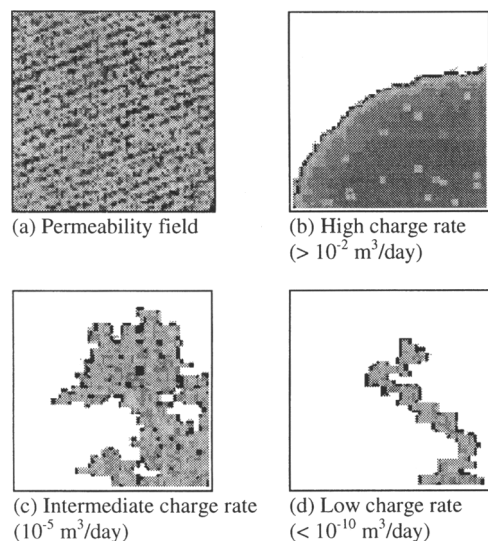
pressure head of the oil phase and the viscous pressure gradient across the pore throat. Therefore, as the injection rate increases, the viscous forces begin to dominate until, as in Fig. 3b, the effective entry pressures approaches zero and a dispersed oil front is observed.

The feeding rates for 'real' systems have been estimated by several authors. For example, Mackenzie *et al.* (1987) estimated the expulsion rates of petroleum from Kimmeridge clay source rocks. For a typical petroleum density of 600 kg m<sup>-3</sup> and a typical source-rock density of 2400 kg m<sup>-3</sup>, they calculated that 19 kg of petroleum per tonne of source rock would be expelled. Most of this oil will be expelled at temperatures between 120–150°C (Cooles *et al.* 1985). If we use burial rates of 35–50 m/Ma (Mann & Mackenzie 1990) and a linear geothermal gradient of 30°C km<sup>-1</sup> (derived from data published by Mackenzie *et al.* 1987), we can estimate that the bulk of the oil (for a single-source horizon) will be expelled over a burial depth of 1 km, corresponding to 20–29 Ma. For a 1 m<sup>3</sup> sample of source rock, this would correspond to a flux rate of  $1.4 \times 10^{-14}$  to  $2.9 \times 10^{-14}$  m<sup>3</sup>m<sup>-2</sup>s<sup>-1</sup>. In another study using less site-specific data, England *et al.* (1987) estimated the flux across the boundary of the source rock to be about  $8 \times 10^{-15}$  to  $8 \times 10^{-14}$  m<sup>3</sup>m<sup>-2</sup>s<sup>-1</sup>.

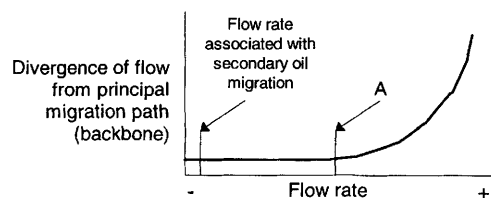
For our model of the inclined, laminated sandstone which has a boundary surface area of 0.6 m<sup>2</sup>, 'realistic' injection rates would therefore be on the order of  $1.2 \times 10^{-9}$  to  $1.6 \times 10^{-9}$  m<sup>3</sup>/day. Present limits to fluid injection in the laboratory are in the order of 1 ml h<sup>-1</sup> ( $24 \times 10^{-6}$  m<sup>3</sup>/day), which would suggest that it may be impossible to ever model secondary oil migration experimentally.

There is, however, a practical limit to how low this injection rate needs to be in order to replicate a realistic oil front. It is related to the threshold pressure of the system and to the ratio of the viscous and capillary forces. This is illustrated in Fig. 4.

At high flow rates where the pressure gradient across the sample is in excess of the threshold pressure, the flow occurs in areas outside of the



**Fig. 3.** Effect on the oil migration flow path of increasing the injection rate.



**Fig. 4.** Relationship between flow rate and flow path divergence.



flow backbone, since the capillary forces are overcome by both the gravitational effects of the oil phase (buoyancy) and, more prominently, the viscous gradient. As the flow rate decreases, the relative balance of the viscous and capillary forces changes from a system which is viscosity dominated to one which is dominated by the capillary forces. At some point (shown as point 'A' in Fig. 4), any further reduction in the injection rate will have no discernible effect on the morphology of the oil front: all flow, subsequent to the first charge, will occur along the backbone of the media.

Quantifying point 'A' is a challenge owing to the heterogeneous nature of carrier systems. Rapport (1955) developed a series of scaling groups for 'use, design and operation of water-oil flow models' which were intended to address this issue. Of interest to this study is the viscous-capillary scaling group, defined by Equation (2):

$$\frac{Q\Delta x\mu}{k_x\Delta y\Delta z\left(\frac{dP_c}{dS_w}\right)} \quad (2)$$

The critical issue is the balance between the threshold pressure and the injection pressure (and hence the flow rate). The precise relationship between these variables can be found both analytically (for homogeneous systems), and through multiphase flow simulations (for heterogeneous configurations). However, it is sufficient to note that if the pressure drop across the sample increases, then the viscous-capillary ratio increases (as shown in Figure 5). This occurs because the flow rate ( $Q$ ) increases proportionally to any increase in the pressure gradient ( $\Delta P$ ) (Darcy's law, Equation 3). Ideally then, experimental work on oil migration should keep

the pressure drop as close as possible to the threshold pressure of the sample.

$$Q_x = -A \frac{k \Delta P}{\mu x} \quad (3)$$

Before looking in more detail at breakthrough pressures, it is worth noting an interesting aspect of flow behaviour at the low flow rates associated with secondary oil migration. At these rates, the flow occurs at hydrostatic equilibrium (as was observed experimentally by Selle *et al.* 1993). From a practical standpoint, this means that both the rate-limiting factor, as well as the dominant control on the trajectory of the migration pathway, is the threshold pressures of the carrier members, not their permeabilities (although there is an implicit relationship between the two).

This can be demonstrated using the model shown in Fig. 3. We ran two cases of the model, both identical except that one model had a permeability distribution multiplied by a factor of 10 over the other. For both models, we ran the simulation through to breakthrough at high and low (realistic) injection rates (rates are shown in Fig. 3). The results are shown in Fig. 6.

The oil saturation profiles for the low and high permeability instances were taken from identical time steps. These eight profiles show that when the injection (feeding) rate decreases towards a value which is represented by point A on Fig. 4, the oil-rock contact volume loses its dependence on the permeability of the medium, and is instead controlled exclusively by the medium's internal threshold pressure field. Note the equivalence between the low and high permeability cases for the low injection rate scenarios.

This can be explained by the carrying capacity of the flow field and the time needed for equi-

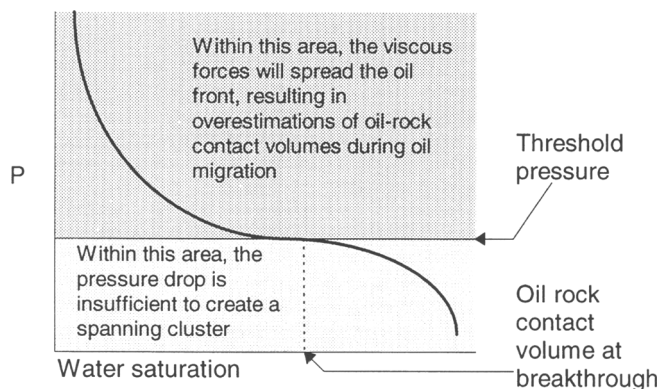
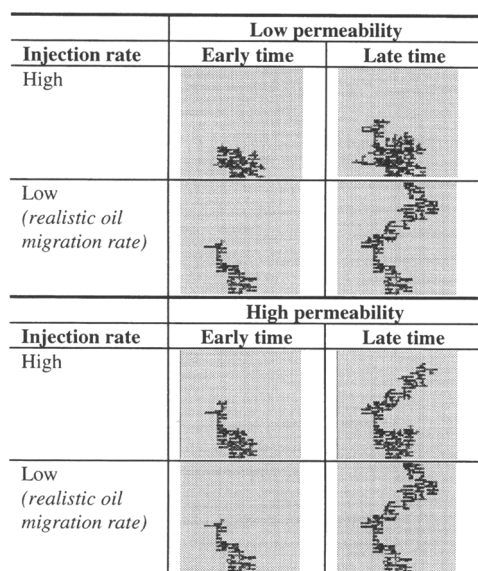


Fig. 5. To simulate oil migration, flow rates must be in accord with the threshold pressure of the sample.



**Fig. 6.** Results of models showing the independence of oil-rock contact volume to permeability, at flow scales associated with secondary migration.

bration to the applied pressure gradient. For a given volume of material, the time taken for a fluid to traverse this volume is a function of its permeability and the applied pressure gradient according to Darcy's law (Equation 3). In the case of the low pressure gradients associated with oil migration, this travel time is very small relative to the Darcy velocity across the sample. The permeability of the sample is therefore insignificant when compared with the effects of the internal capillary pressure field.

### Assessment of breakthrough pressures and associated oil-rock contact volumes

As the oil phase moves from the inlet to the outlet of a given carrier volume, it invades all the pores which have entry pressures equal to or lower than the pressure head of the oil stringer or oil column. The height of an oil column at breakthrough was defined by Berg (1975) as 'the vertical height that must be attained for the buoyant force to equal the internal-pressure gradient (the critical height,  $z_c$ , for migration)':

$$z_c = 2\gamma \frac{\left(\frac{1}{r_i} - \frac{1}{r_p}\right)}{g(\rho_w - \rho_o)}. \quad (4)$$

Calculating this critical height using this equation requires an estimate of the largest pore

throat radius within the migration pathway, which is not easily quantified. Schowalter (1979) recognized this and suggested that all the information we need for estimating migration potential can be gathered from mercury intrusion experiments. He states: 'if one can determine the pressure required to establish a connected hydrocarbon filament through the largest interconnected water-saturated pore throats, one can calculate the vertical hydrocarbon column required to migrate hydrocarbons'. In his work, he used a nitrogen-water system in which nitrogen was displaced through a water-filled rock sample under a confining pressure. The nitrogen pressure was increased and the amount of effluent at the sample outlet was monitored. The nitrogen pressures were then converted to an equivalent mercury pressure by scaling the interfacial tensions and contact angles of the two fluids.

He ran this experiment on a range of samples, which included sandstones, silty shales and chalks (some of his results are shown in Fig. 7). He found that the non-wetting phase saturation needed to establish a connected filament across the length of the samples ranged from 4.5% to 17%, with an average value of 10%. Based on these experiments, Schowalter suggested that 'displacement [threshold] pressures could be estimated from standard mercury capillary pressure curves by determining the mercury phase pressure on the capillary curve at 10% mercury saturation'.

Expressed in terms of the arguments of this paper, Schowalter found that the oil-rock contact volume necessary to create a spanning cluster through a range of lithotypes, was on average 10% of the effective pore volume.

Swanson (1977) showed that 'the point on a mercury injection curve that represents a continuous, well-interconnected pore system through the rock' corresponds with the apex of the hyperbola of a log(pressure)-log(saturation) plot. Pittman (1992) showed that the apex of the hyperbola can be more accurately identified by plotting the ratio of Hg saturation/capillary pressure against the corresponding Hg saturation. The rationale for this approach is that the threshold pressure corresponds to the inflection point at which the curve becomes convex upwards (Katz & Thompson 1987), and by plotting the data in this manner, the inflection point will manifest itself as the apex of the hyperbola. An example of this plot is shown in Fig. 8, where we applied this technique to capillary pressure data derived from North Sea core plugs.

For these samples, the intruding phase saturations at breakthrough varied between roughly

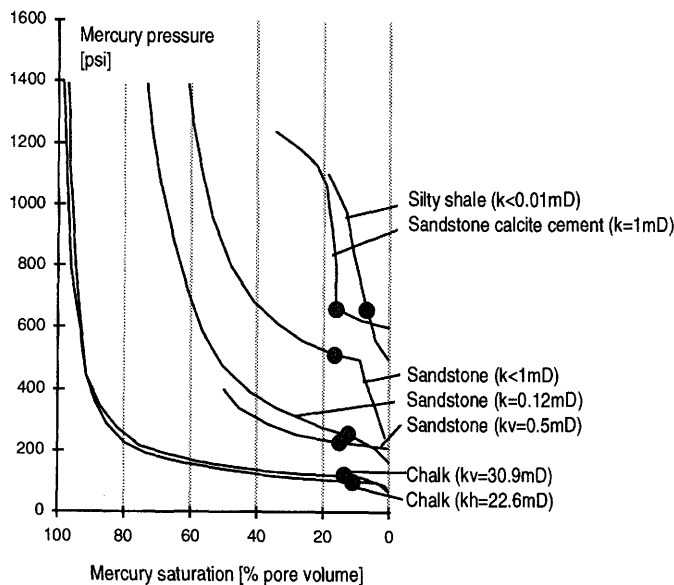


Fig. 7. Threshold pressures and saturations (shown by the large circles) for a range of rock types (Schowalter 1979).

15% and 40%. Pittman (1992) conducted a similar analysis for 196 sandstones from 14 formations in which the porosities and permeabilities of his data set ranged from 3.3 to 28.0% and 0.05 to 998 mD, respectively. He reports a mercury saturation of 36% corresponding to the mean apex. Furthermore, a regression on his 196 samples revealed an empirical relationship between the pore size at the apex and permeability:

$$\log(r_{\text{apex}}) = -0.226 + 0.44 \log(k). \quad (5)$$

Using Equation (6) (Washburn 1921), which relates pore radius to capillary pressure, we can extend Pittman's empirical relationship to equate permeability with threshold pressure (Equation 6a). For water-wet carriers, the cosine of the contact angle is 1.

$$P_c = \frac{-2\gamma \cos \theta}{r_{\text{apex}}} \quad (6)$$

$$P_c = \frac{-2\gamma \cos \theta}{10^{-0.226+0.446 \log(k)}} \quad (6a)$$

Figure 9 shows the graphical relationship between  $k$  and  $P_c$  for a range of interfacial tensions using Equation (6a).

The range of non-wetting phase saturations at breakthrough (i.e. oil-rock contact volumes required for oil migration) in the examples given here have covered a pore volume range of

4.5% to nearly 45%. This large variation is due to the heterogeneity within the samples as well as the orientation of the sedimentary fabric relative to the direction of the net migration vector. Since it is the saturation at breakthrough which defines the oil-rock contact volume during secondary migration, we need to look more closely at how the heterogeneity of the medium affects the breakthrough pressures. In a heterogeneous environment, in the presence of migration baffles

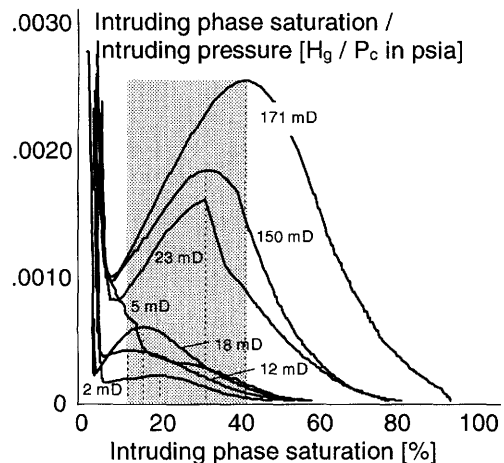
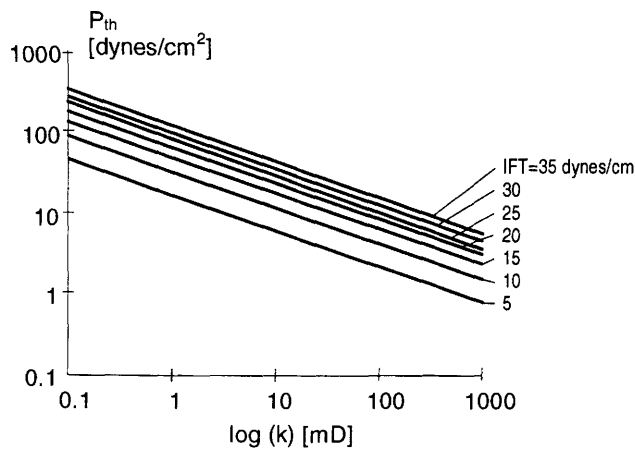


Fig. 8. Demonstration of Pittman's (1992) technique applied to six North Sea samples.





**Fig. 9.** Relationship between permeability ( $k$ ), interfacial tension ( $IFT$ ) and threshold pressure ( $P_{th}$ ) using Pittman's (1992) empirical equations and the Washburn (1921) equation. 1 dyne/cm<sup>2</sup> is equivalent to  $10^{-6}$  bars.

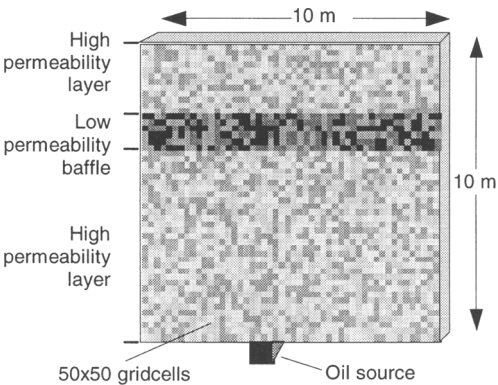
and barriers, the oil–rock contacts are dynamic, reflecting the transient nature of the pressure head of the oil phase. The models below can be used to demonstrate this dynamism.

**Threshold saturations in a heterogeneous system**

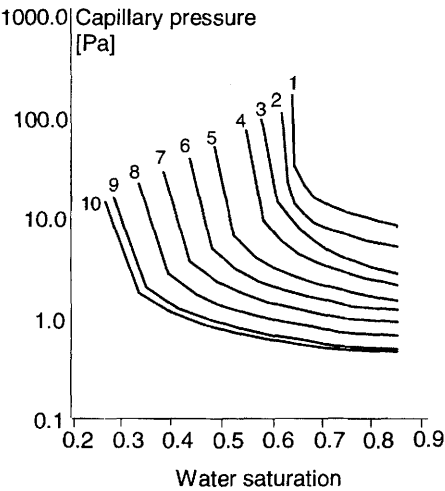
*The effects of a heterogeneous threshold pressure field*

To evaluate the effects of a heterogeneous internal capillary pressure field (i.e. heterogeneous pore throat distributions), we simulated the upward migration of oil through a high permeability carrier, beneath a baffle. The configura-

tion of the model is shown in Fig. 10, and the properties of each of the ten discrete units represented by this model are shown in Fig. 11 and in Table 1. Oil is slowly injected at the base of the model at a rate of  $10^{-14}$  m<sup>3</sup>/m<sup>2</sup>/s, which for a model with a base surface area of 100 m<sup>2</sup>, corresponds to an injection rate of  $10^{-12}$  m<sup>3</sup> s<sup>-1</sup>, which is approximately  $10^{-11}$  m<sup>3</sup>/day. This value is well within the capillary-dominated range shown in Fig. 5. England *et al.* (1987) state that 'experiments with oil-water mixtures in rocks have shown that at capillary numbers greater than  $10^{-4}$ , viscous forces become important', where the capillary number is defined by



**Fig. 10.** Configuration of the model. Ten discrete units were modelled, which are represented in this diagram via the ten grey tones which comprise the figure.

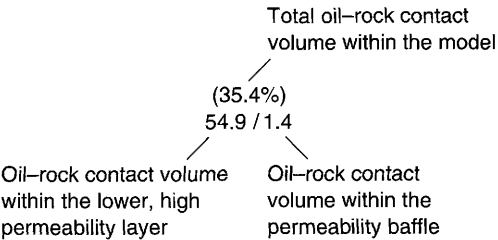


**Fig. 11.** Capillary pressure curves used in the model shown in Fig. 10.

**Table 1.** Permeability, relative permeability end points and index of the  $P_c$  curves used for each of the ten units represented in the model

| Unit number | $k$ (mD) | $S_{wc}$ | $S_{wor}$ | $P_c$ curve |
|-------------|----------|----------|-----------|-------------|
| 1           | 0.01     | 0.640    | 0.85      | 1           |
| 2           | 0.1      | 0.620    | 0.85      | 2           |
| 3           | 1        | 0.580    | 0.85      | 3           |
| 4           | 2        | 0.550    | 0.85      | 4           |
| 5           | 5        | 0.485    | 0.85      | 5           |
| 6           | 10       | 0.435    | 0.85      | 6           |
| 7           | 20       | 0.385    | 0.85      | 7           |
| 8           | 40       | 0.336    | 0.85      | 8           |
| 9           | 80       | 0.286    | 0.85      | 9           |
| 10          | 100      | 0.270    | 0.85      | 10          |

sample volume. The dotted line is included for reference purposes and delineates the base of the baffle. The numbers on the figure refer to the oil–rock contact volumes within the overall model, the baffle, and the lower unit, respectively. For example, for time step K, the numbers are:

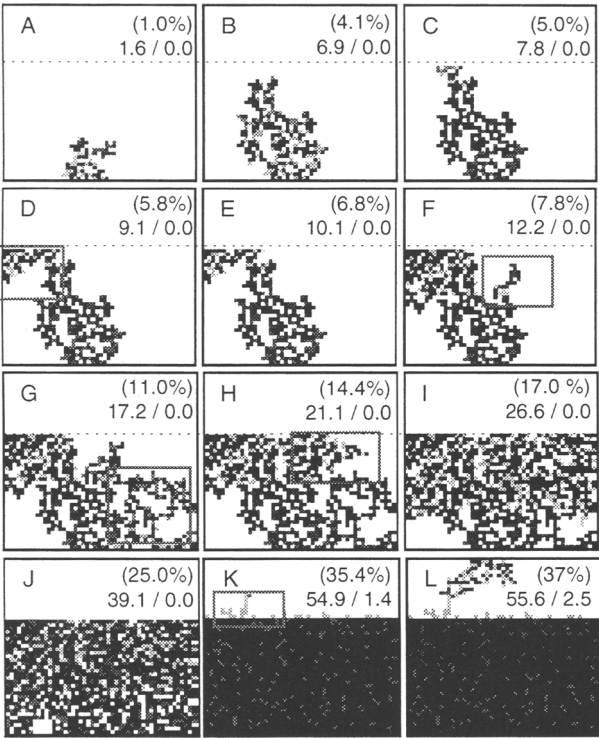


Equation (7). Our injection rates are therefore well within the capillary dominated flow regime. No relative permeability or capillary pressure hysteresis was modelled.

$$C_a = \frac{\mu q}{\gamma} \tag{7}$$

Figure 12 shows the oil saturation profiles for the time steps leading to the breakthrough of the

At time step A we see the early development of the oil stringer. Each of the grid blocks it has saturated so far, correspond with the grid cells which neighbour the injection point, and have the lowest threshold pressures of all the grid blocks connected to the injector. In time steps B–C, we see the continued development of the stringer, as buoyancy drives it upwards towards the baffle.



**Fig. 12.** Oil saturations from the three-layer model.

At time step D there is sufficient potential energy within the oil stringer to overcome the lowest threshold pressure in the adjacent cells, which occurs in the inner box on the figure. It is only slightly coincidental that this cell is at the top of the oil cluster, but it coincides with the area in which both the oil saturations are highest (and therefore the oil mobilities are highest) and the magnitude of the buoyancy force is at its peak.

For time steps D to E the stringer has reached a critical point: there are no cells adjacent to any oil-saturated cells which have threshold pressures lower than the pressure head corresponding to the amount of oil which has been injected. At this point, the pressure within the oil phase increases uniformly throughout the stringer due to Pascal's principle, driving the oil pressure up the capillary pressure curve resulting in higher saturations throughout the stringer. This excess pressure within the oil phase is finally released at the top of the stringer, delineated by the inner box on time step F.

Note that at time step E the saturated area beneath this box has remained virtually unchanged since it was first spanned at time step B. This observation supports the idea that once a given volume is spanned, the oil saturation within that volume will remain constant. In other words, as long as the threshold pressure within this smaller region is the same or lower than the threshold pressure of the larger volume of which it is a part, the oil saturation within this smaller volume will never increase beyond the saturation corresponding to its threshold pressure.

At time step G, The oil finds it easier to expand downwards rather than continue its expansion at the top of the stringer. This has an interesting consequence for predicting migration pathways. What has happened is that, as the pressure head increases, this pressure increase occurs uniformly throughout the oil phase. As a result, it is not necessarily at the top of the stringer where the expansion will occur. As we see in time steps G and H, the oil finds it easier to expand downwards through the carrier, rather than accumulate beneath the capillary baffle.

At time steps J through to K, the oil accumulates beneath the baffle. It is not until time step K that the oil has sufficient potential energy (and sufficient pressure head) to overcome the capillary entry pressure of the baffle. At this point the oil rapidly breaks through the baffle and begins to repeat its migration and expansion into the upper sandstone unit.

Based on this model we can infer some general characteristics of migration:

1. In a hydrostatic environment, the net migration vector is directed upwards, although locally the migration can be directed downwards if a preferential pathway exists in that direction.
2. As long as the threshold pressure within a sub-set of the carrier is lower than any upstream threshold pressures, the oil saturations within this sub-set will remain fixed at the saturation corresponding to its threshold pressure.
3. For a given pressure head, all pores which have threshold pressures lower than the oil pressure head, and are in turn connected to similar pores which are ultimately connected to the source, will 'see' the oil phase during secondary migration.
4. The oil-rock contact volumes will be highest beneath a seal (or baffle) and lowest in the baffle itself.

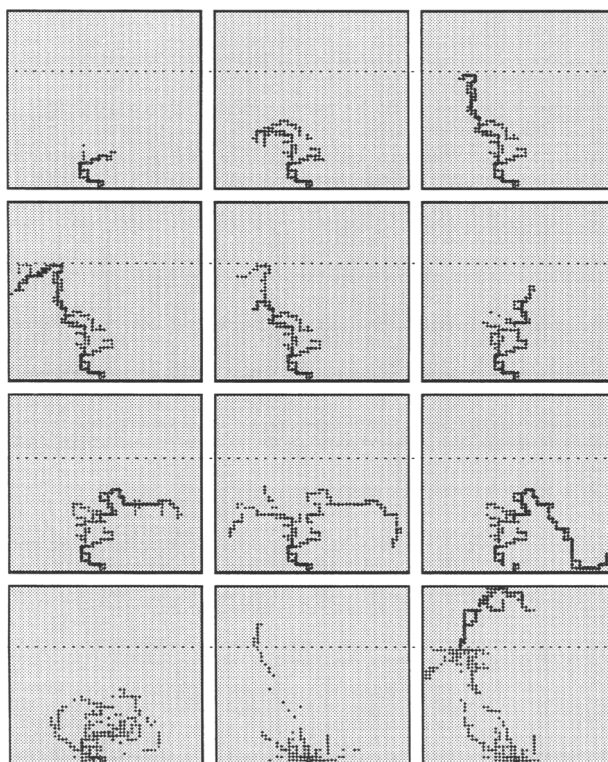
To help visualize where the oil flow is occurring, we can plot the maximum oil flux for the same model (Fig. 13). Each plot in the diagram represents a single time step. In plotting the data in this way, we can more easily visualize the migration trajectory for a given time step (in which a time step is equivalent to the instantaneous oil pressure head).

The first three plots show the path of least resistance (the flow backbone) through the lower unit. In the second row, we see that the oil cannot break through the baffle with the available pressure head, so it finds an alternative pathway towards the right side of the model. During this stage, the buoyancy force is accumulating within the oil phase until, as is shown in the second last plot, the baffle is breached. The last plot in the series represents the flow backbone through the entire modelled formation. Subsequent flows will be concentrated along this path.

Note that if we were to plot the volume of oil which each grid cell 'sees', or if we were to associate an age to this oil, then we would find that the flow backbone will see the oil at all stages of charging, whereas areas outside of the backbone will see only the earliest charges. This has important implications for the interpretation of geochemical indicators of petroleum history.

Based on plotting the flow backbone for each time step, we can infer the following:

1. The migration trajectory is a function of the instantaneous pressure head of the oil phase;
2. The oil-rock contact volume for the first charge will be high, until the threshold pressure is reached for the carrier volume;



**Fig. 13.** Plots of the paths of maximum fluid flux for nine timesteps (migration trajectories). The horizontal dotted lines represents the base of the low permeability baffle. Compare these flux profiles with the saturation profiles in Fig. 12.

3. Once the threshold pressure is reached (i.e. the system percolates), flow will be concentrated along the backbone, and if the oil phase is continuous along this backbone, all flow will occur via this pathway.

In this section we have seen how oil migration (and oil–rock contact volume) is controlled by the instantaneous pressure head within the oil phase and by the configuration of the threshold pressure field. In the next section we evaluate the effects of sedimentary structure on oil–rock contact volume using a high resolution model of a bioturbated silty sandstone.

### *The effects of sedimentary structure*

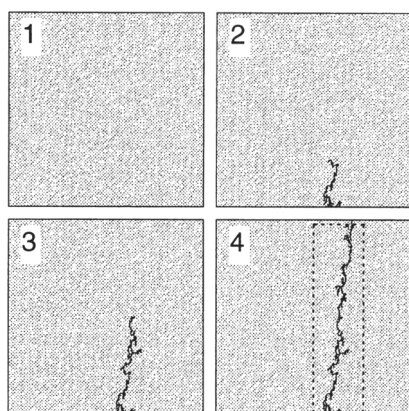
The models in the previous section have shown the dynamic effects that a static capillary threshold field can have on both migration trajectories and oil–rock contact volumes during a gravity–destabilized invasion of a non-wetting oil phase into a water-wet carrier. These models are

useful in their ability to increase our understanding of how the contact volumes will change according to both the timing of the oil charge (suggesting that the earlier charges will see the greater volume of rock) and according to the instantaneous pressure head (reflected in the unstable position of the backbone in Fig. 13).

Earlier in this paper we showed that injection experiments have yielded oil–rock contact volumes of between 4% and 45% at breakthrough. In this section, we examine the effect of sedimentary structures on oil–rock contact volumes by simulating the migration of oil through a bioturbated, heterolithic, shallow marine facies. These simulations will provide insights into what might occur to a fluid front during injection experiments, and will assist us in constraining the contact volumes for carrier media.

A heterogeneous capillary threshold field in itself is insufficient to create large contact volumes during secondary migration. Figure 14 shows the results of a numerical invasion percolation-type simulation of a  $600 \times 600$  grid. The





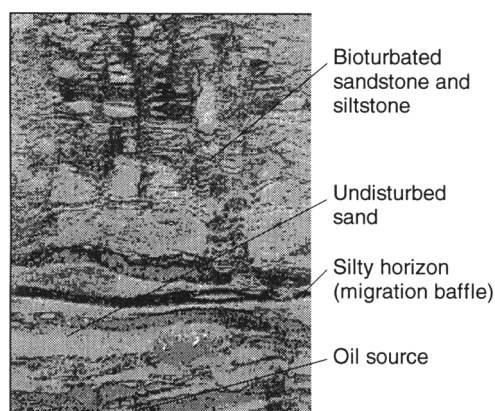
**Fig. 14.** Model to demonstrate how the lack of sedimentary structure will result in a very low oil-rock contact volume, despite having a wide range of internal capillary pressure values.

details of these types of models have been published by others and will not be reproduced here. Hirsh & Thompson (1995), and Wagner *et al.* (1997) give more information on invasion percolation models applied to secondary oil migration modelling.

Each cell in the model was randomly assigned a normalized capillary threshold value ranging from 0 to 100, drawn from a log-normal distribution. In a field of this type, where no sedimentary structure exists, we see that the contact volume of the invading oil phase is very low. In fact, this type of modelling led Hirsch & Thompson (1995) to conclude that 'field-scale saturations in secondary migration can be on the order of 1% or less [of the carrier pore volume]'.

If we look at Fig. 14(4), the dotted box represents the volume of material *necessary* for the system to percolate. That is to say, there is a required volume necessary in order for migration to occur under a given pressure regime. Any material outside of this region is superfluous and only serves to decrease the calculated oil-rock contact volume by increasing the amount of non-contacted material. Hirsch & Thompson (1995) described the effects of sample size on saturation as an 'inverse square root dependence of saturation on sample size [which] indicates that the invasion process is a percolation process and that the fractal dimension of the percolation cluster is 2.5'.

This type of modelling is valuable for quantifying the oil-rock contact volumes associated with secondary oil migration. However, the lack of any sedimentary and capillary architecture within the models can lead to an under-estimation of the contact volumes.



**Fig. 15.** Bioturbated, shallow-marine, heterolithic facies. The width of the sample was 10 cm.

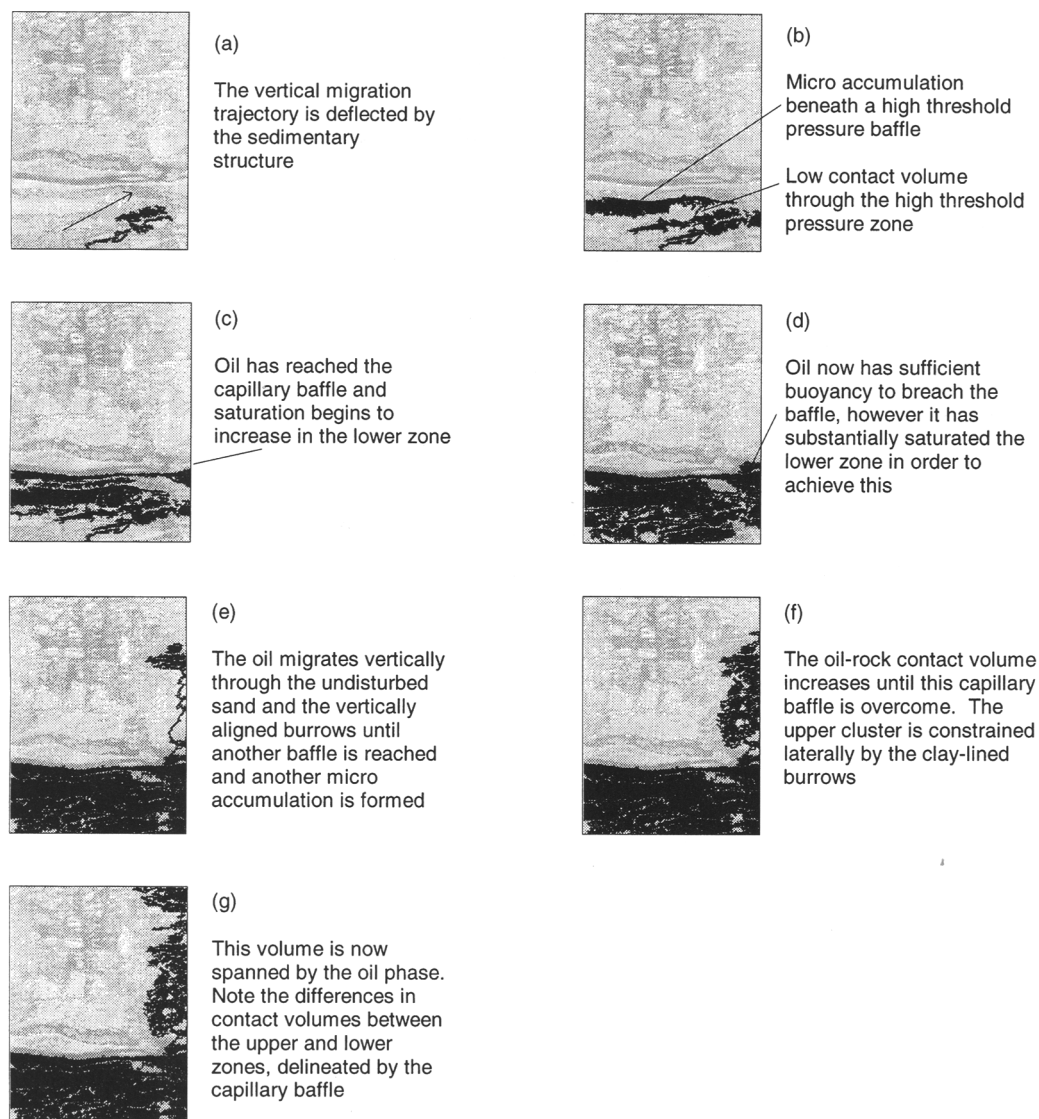
To demonstrate this, we applied a modified invasion percolation technique to the lithotype represented in Fig. 15. This figure shows an image of a section of core taken from a bioturbated interval, deposited in a shallow marine environment. The grain sizes range from clay-silt to medium sand, and the sample was chosen based on its wealth of internal structures which may play a role in the secondary migration process.

To preserve the high resolution nature of the data, we converted a core photograph to a grey-scale image and assigned capillary threshold values to our model based on these grey-scale values. The more clay-rich sediments are darker in colour than the lighter coloured sands, so grey scale was assumed to represent the capillary pressure field. A numerical oil source was placed in the middle of the lower boundary and oil was allowed to percolate until it spanned the model in a vertical direction. Lateral no-flow boundary conditions were applied. Model dimensions were  $760 \times 1024$  cells, and the results are shown in Fig. 16.

In Fig. 16a, we see how the early migration through the volume is affected by the inclined sedimentary structure of the lower sand zone, resulting in a deflection of the migration trajectory towards the right. This continues until it reaches the right margin of the model where, due to the no-flow boundary conditions, the flow is redirected back towards the centre.

This is precisely the type of flow which will occur in most core floods. In most fluid flow experiments (both numerical and physical), boundary conditions will play a significant role and including their effects in our interpretation is critical. In this particular case, had we been





**Fig. 16.** Series of saturation profiles of oil migrating through the bioturbated heterolithics shown in Fig. 15.

evaluating a formation unconstrained by boundary conditions, it is unlikely that the contact volumes within the volume represented by Fig. 16a would increase much beyond what is shown in Fig. 16a, since the flow would have been diverted into neighbouring volumes by the sedimentary structures. Such limitations aside, these types of models are still extremely valuable since they provide an estimate of the *maximum* oil-rock contact volume which is likely to occur in a given rock volume.

In Fig. 16b the oil continues to migrate, and

the buoyancy force within the oil stringer is sufficient to breach a small baffle and create a micro-accumulation beneath a larger one. Note the low contact volume within the baffle, suggesting that contact volumes may be much lower in mudrock or heterolithic sequences than in contrasting sandstone sequences were all migration to occur via pores (as opposed to fractures). This detailed visualization of oil migration in a realistic rock pattern helps resolve some of the previously conflicting views on oil-rock contact volumes.

By Fig. 16c, the oil has reached the laterally extensive capillary baffle and an accumulation begins to form. Had this baffle not existed, the contact volume within the lower portion of the model would remain essentially the same as that in Fig. 16b. The presence of this seal causes the oil saturation to increase until, in Fig. 16e, the seal is fully breached and the oil can continue its upward vertical migration.

In Fig. 16e–f, the oil migrates relatively unimpeded through the upper zone, since the net to gross ratio within this area is much higher than the lower zone and the presence of the bioturbation creates focused vertical conduits by imposing lateral barriers to migration. By Fig. 16g, the volume is fully spanned. The oil–rock contact for the entire volume at this time step is approximately 30%.

## Discussion

In this paper we have seen two different approaches to modelling secondary oil migration: finite difference simulation and the simulation of an invasion percolation network. The finite difference simulator assumes that the oil phase is continuous, and therefore any pressure increases within the oil phase occur ubiquitously throughout the oil phase. As was mentioned in our conceptual view on how oil migrates, this may be an inapplicable assumption for modelling oil migration, since the continuity within the oleic phase will probably occur only at the most upstream limits of the otherwise disconnected oil ganglia.

These models were of 2D systems. Percolation theory tells us that moving to 3D systems will result in much lower contact volumes, since a much lower percentage of oil-filled pores is necessary to create a spanning cluster across a given volume (Chandler *et al.* 1982). It is therefore logical to conclude that the figures we have quoted for contact volumes represent the *highest* values which one is likely to encounter in each of the systems given. That in itself is not a limitation, since it is perhaps more meaningful to generate detailed 2D models than to compromise details for computing efficiency in the creation of 3D models having much lower resolutions.

There can never be a single value for an oil–rock contact volume associated with migration through a carrier, since there is no such thing as a single carrier type. Carrier systems were derived from stacked depositional sequences which create a heterogeneous and site-specific mix of flow units and lithotypes. Therefore, each carrier system needs to be evaluated indi-

vidually. A possible course of action would be to look at the carrier properties of different depositional systems and evaluate their threshold pressures and associated oil–rock contact volumes. For example, one might derive these parameters for a shallow marine or mixed aeolian–fluvial system. We have already started doing this for the shallow marine case.

In any system, the boundary conditions and the areal extent of lateral baffles and seals will strongly influence the oil–rock contact volumes. We therefore identify two systems: those with laterally continuous barriers and those without. If a barrier is continuous then we can associate the 'vertical hydrocarbon-height required to breach a capillary seal' (Equation 4) with a measure of contact volume. Otherwise, a discontinuous baffle will deflect the migration and will increase the contact volume proportional to the distance deflected.

## Conclusions

In this paper we have shown that the oil–rock contact volumes associated with secondary oil migration are a function of the heterogeneity of the capillary threshold field within the lithofacies which collectively form a carrier sequence. Due to the extremely low viscous–capillary ratio, the gravity-destabilized invasion of the oil phase into a water-wet carrier occurs at hydrostatically equilibrated conditions. As a consequence of this, the oil phase will always try to minimize its potential energy by choosing the path of least capillary resistance through a given carrier volume. Once this path is found, all subsequent oil flow will occur along this path.

Saturations may increase beyond this minimum pathway where areally extensive capillary baffles or seals exist. If these barriers are large enough in their extent, economic accumulations may develop. Independent of their economic potential, all baffles and seals force the oil into pores outside of the path of least resistance which, over the course of the feeding cycle, may have the effect of increasing the oil–rock contact volume for the initial charge.

The large range of saturations required to migrate oil through a given volume, when measured experimentally, covers at least one order of magnitude (core plug values between 4% and 45% were quoted in this paper). This range is attributed to the differences in sedimentary fabric between the samples and how this fabric is oriented relative to the migration trajectory.

A general observation that can be made is that lower permeability rock units will generally have very low oil–rock contact volumes. High permeability units can have low or high oil–rock contact volumes depending on the spatial distribution of baffles.

Detailed simulation of oil invasion into a range of lithotypes can provide insights into necessary contact volumes through larger-scale carrier sequences. By building detailed carrier models based on these lithotype-specific carrier models, and by using the threshold pressure and corresponding contact volume information derived from the types of models shown in the last section, we will be able to predict more accurately the elusive oil–rock contact volume (and therefore migration efficiency) associated with oil migration through a range of carrier types.

We would like to thank Steve Larter of Newcastle University for discussions on source rock feeding rates, Tom Manzocchi of the Fault Analysis Group at Liverpool University for useful debates on network modelling, and Malcolm Arnot of Heriot-Watt University for discussions on the effects of sedimentary structures. We also gratefully acknowledge funding support provided by the EC under Joule II contract JOU2–CT93–0368 and Joule III contract JOF3–CT95–0014.

## Nomenclature

|            |                                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------|
| $A$        | Area (length <sup>2</sup> )                                                                             |
| $C_a$      | Capillary number                                                                                        |
| $g$        | Acceleration due to gravity (length time <sup>-2</sup> )                                                |
| $k$        | Permeability (area <sup>2</sup> )                                                                       |
| $P$        | Pressure (pressure)                                                                                     |
| $P_c$      | Capillary pressure (pressure)                                                                           |
| $q$        | Darcy flux per unit cross-sectional area (length <sup>3</sup> length <sup>-2</sup> time <sup>-1</sup> ) |
| $Q$        | Flow rate (length <sup>3</sup> time)                                                                    |
| $r_t, r_p$ | Radius of the pore throats and pores bodies (length)                                                    |
| $S_w$      | Water saturation (%)                                                                                    |
| $S_{wc}$   | Connate water saturation (%)                                                                            |
| $S_{wor}$  | The water saturation which corresponds to the residual oil saturation (%)                               |
| $x, y, z$  | Direction (with $z$ being vertical)                                                                     |
| $z_c$      | Critical height for migration (length)                                                                  |
| $\gamma$   | Interfacial tension (force length <sup>-1</sup> )                                                       |
| $\theta$   | Fluid contact angle (radians)                                                                           |
| $\mu$      | Viscosity (mass length <sup>-1</sup> time <sup>-1</sup> )                                               |
| $\rho$     | Density (mass length <sup>-3</sup> )                                                                    |

## Appendix

Empirical relative permeability and capillary pressure functions (Ringrose & Corbett 1994).

Connate water saturation:

$$S_{wc} = 0.6 - 0.165 \log_{10}(k_{abs})$$

Water saturation at residual oil:

$$S_{wor} = \text{constant} = 0.85$$

Effective water saturation:

$$S_e = \frac{S_w - S_{wc}}{S_{wor} - S_{wc}}$$

Porosity:

$$\phi = 0.05 \log_{10}(K_{abs}) + 0.076$$

Capillary pressure (bars):

$$P_c = 3.0 S_e^{-2/3} \sqrt{\frac{\phi}{k_{abs}}}$$

Relative permeabilities:

$$k_{rw} = 0.3 S_e^3$$

$$k_{ro} = 0.85(1 - S_e)^3$$

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